

Discrete Probability distributions: Uniform and Binomial distributions

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Some Probability Distributions

Discrete Probability distributions

- 1 Discrete uniform distribution.
- 2 **Binomial Distributions.**
- 3 Hyper geometric distribution.
- 4 **Poisson Distribution.**

Continuous Probability Distributions

- 1 Continuous Uniform Distribution.
- 2 Exponential distribution.
- 3 **Normal distribution**

Discrete uniform distribution

- The simplest of all discrete probability distributions is one where the random variable assumes each of its values with an equal probability. Such a probability distribution is called a **discrete uniform distribution**.

Definition

If the random variable X assumes the values $1, 2, \dots, n$, with equal probabilities, then the discrete uniform distribution is given by

$$f(X_i) = \frac{1}{n}, \quad X_i \in \{1, 2, \dots, n\}.$$

- The moment generating function of the binomial distribution is given by

$$G_X(t) = E(e^{tX}) = \frac{e^t(1 - e^{nt})}{n(1 - e^t)}.$$

cont...

Proposition

The **mean** and **variance** of the discrete uniform distribution are given

$$f(X_i) = \frac{1}{n}, \quad X_i = 1, 2, \dots, 6.$$

$$E(X) = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1+n}{2}$$

$$\sigma_x^2 = \frac{1}{n} \sum_{i=1}^n (X_i - E(x))^2 = E(x^2) - E^2(X) = \frac{n^2 - 1}{12}.$$

Example

When a fair die is tossed, each element of the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$ occurs with probability $\frac{1}{6}$.

Example with answer

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Therefore, we have the uniform distribution, with

$$E(X) = \mu = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5;$$

$$\text{Variance} = \sigma_x^2 = \frac{1}{n} \sum_{i=1}^n (X_i - E(x))^2 = E(x^2) - E^2(X) = 2.92.$$

Special discrete probability distributions

Binomial distribution

- **Binomial experiment** is an experiment with a fixed number of independent trials.
- Each trial can only have two outcomes, or outcomes which can be reduced to two outcomes.
- The probability of each outcome must remain constant from trial to trial.
- **Binomial distribution** is the distribution of the outcomes of a binomial experiment with their corresponding probabilities.

Binomial distribution

- Let consider an experiment with n **independent trials** and **only two possible outcomes**.
- We call one of the outcomes **successful outcome** and the probability of its occurring is $P(s) = p$.
- The other outcome is called **fail outcome** and the probability of its occurring is $P(f) = q = 1 - p$.

The probability for getting exactly k successes in n trials is

$$P_k = B(k, n, p) = B(n, p) = \binom{n}{k} p^k q^{n-k} \text{ and } k = 0, 1, 2, \dots, n.$$

Then the pdf of X is given by $f(x) = P(X = x) = \binom{n}{k} p^k q^{n-k} \text{ and } k = 0, 1, 2, \dots, n$

We write $X \rightsquigarrow \text{Bin}(n, p)$ and read “ X is a random variable following the binomial distribution with parameter n and p ”.

Example: The probability that a person supports party A is 0.6. Find the probability that a randomly selected sample of 8 voters, there are:

- a) Exactly 3 who support party A.
- b) More than 5 who support party A

Solution: Consider “support of party A to be success”. Then $p=0.6$ and $q=1-0.6=0.4$.

Let X be the r.v. “the number of party A supporters”

$$X \sim \text{Bin}(n, p)$$

$$X \sim \text{Bin}(8, 0.6) \text{ and its pdf is given by } f(x) = P(X = x) = \binom{n}{k} p^k q^{n-k} \text{ and } k = 0, 1, 2, \dots, n$$

$$f(x) = P(X = x) = \binom{8}{k} (0.6)^k (0.4)^{8-k} \text{ and } k = 0, 1, 2, \dots, 8$$

Thus

$$f(x) = P(X = x) = \binom{8}{k} (0.6)^k (0.4)^{8-k} \text{ and } k = 0, 1, 2, \dots, 8$$

Thus

a) k=3

$$P(X=3) = \binom{8}{3} (0.6)^3 (0.4)^{8-3} = 0.124$$

$$P(X > 5) = P(X=6) + P(X=7) + P(X=8)$$

$$= \binom{8}{6} (0.6)^6 (0.4)^{8-6} + \binom{8}{7} (0.6)^7 (0.4)^{8-7} + \binom{8}{8} (0.6)^8 (0.4)^{8-8}$$

$$= \binom{8}{6} (0.6)^6 (0.4)^2 + \binom{8}{7} (0.6)^7 (0.4)^1 + \binom{8}{8} (0.6)^8 (0.4)^0$$

$$= \mathbf{0.315}$$

Expectation value and variance.

If the r.v X is such that $X \sim \text{Bin}(n, p)$, then $E(X) = np$ and $\text{Var } X = npq$ where $q = 1 - p$

Example: If the probability that it will be a fine day is 0.4. Find the expected number of fine days in a week end and the standard deviation.

Solution

Let “fine day” be “success”. Then, $p=0.4$ and $q=0.6$.

Let X be an r.v. “number of fine days in a week “.Then $X \sim \text{Bin}(n, p)$, where $n=7$ and $p=0.4$.

Hence $X \sim \text{Bin}(7, 0.4)$

a) $E(X) = np = 7 * (0.4) = 2.8$

b) $\text{Var } X = npq = 7 * (0.4) * (0.6) = 1.68$
 $\Rightarrow \sigma = \sqrt{\text{Var } X} = \sqrt{1.68} = 1.2961$

Mode

Let the r.v. X be such that $X \sim \text{Bin}(n, p)$. To compute the mode of the probability distribution of X , we only consider the values of X close to $E(X)$.

Example:

If $X \sim \text{Bin}(10, 0.45)$, find the mode of the probability distribution of X .

|Solution: Here X has pdf as follow: $f(x) = P(X = x) = \binom{n}{k} p^k q^{n-k}$ and $k = 0, 1, 2, \dots, n$

So that $P(X = x) = \binom{10}{k} (0.45)^k (0.55)^{10-k}$ and $k = 0, 1, 2, \dots, 10$

But $E(x) = np = 10 \cdot 0.45 = 4.5$

Now $P(X = 3) = \binom{10}{3} (0.45)^3 (0.55)^{10-3} = 0.1664$

$P(X = 4) = \binom{10}{4} (0.45)^4 (0.55)^{10-4} = 0.2383$

$P(X = 5) = \binom{10}{5} (0.45)^5 (0.55)^{10-5} = 0.2340$

$P(X = 6) = \binom{10}{6} (0.45)^6 (0.55)^{10-6} = 0.1595$

Here X with the highest probability is 4. Therefore, the mode of X is 4.

Binomial Examples

Example 1: Fair coin tossed 5 times. Probability of exactly 3 tails?

$$n = 5, p = \frac{1}{2}, x = 3 \Rightarrow P(X = 3) = \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{10}{32} = \frac{5}{16}$$

Example 2: Multiple-choice test with 20 questions, 5 choices each. Student guesses randomly.

- $n = 20, p = 0.2$
- Probability of 8 correct answers:

$$P(X = 8) = \binom{20}{8} (0.2)^8 (0.8)^{12} \approx 0.0222$$

Example 3: Mean and variance for number of heads in 3 coin flips:

$$\mu = np = 3 \cdot \frac{1}{2} = 1.5, \quad \sigma^2 = npq = 3 \cdot \frac{1}{2} \cdot \frac{1}{2} = 0.75$$